Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption
One potential explanation for the website's connection timeout error message is:A denial of service (DoS) attack, specifically a SYN flood attack. The logs show that: The IP address 203.0.113.0 sent more than 130 TCP SYN requests to the web server at 192.0.2.1 in about 48 seconds. This IP address is outside the 198.51.100.0/24 range used by employee computers. At first, employees could load the sales page and received HTTP 200 OK responses. As the SYN requests continued, one employee received a 504 Gateway Time-out error, and others had their connections reset. After about 21 seconds, the log shows only SYN requests from 203.0.113.0, with no replies from the server. This event could be:A SYN flood attack. A large number of SYN requests came from one IP address, so this looks like a DoS attack and not a DDoS attack, which would involve many sources.

Section 2: Explain how the attack is causing the website to malfunction
When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake: 1.The visitor’s device sends a SYN packet to the web server to ask to connect. 2. The web server replies with a SYN-ACK packet to agree to the connection and reserves resources for the final step. 3.The visitor’s device sends an ACK packet to confirm, and the connection is established. Explain what happens when a malicious actor sends a large number of SYN packets all at once:The server has to use resources to process each SYN request and may hold connections open while it waits for the final ACK. If too many requests arrive at once, the server runs out of resources. It then becomes slow or stops responding, and real visitors cannot connect. Explain what the logs indicate and how that affects the server:The logs indicate that the attacker at 203.0.113.0 completed one handshake and then kept sending SYN requests to

port 443 without completing another one. Normal traffic worked at first, but the server became overwhelmed as the attack continued. Packet 77 shows a 504 Gateway Time-out error for an employee, and several employees had their connections reset. By the end of the log, the server was no longer sending any responses, so employees could not reach the sales webpage. This could cause lost sales because employees could not look up vacation packages for customers.

I took the server offline temporarily so it could recover and blocked the IP address at the firewall. This may only be a short-term fix because the attacker can spoof other IP addresses. To help prevent this in the future, the company could enable SYN cookies, rate limit SYN requests, and use an intrusion prevention system.